

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16851	
Total Threats Mitigated	56	Total Packets Analyzed		8707188298	
High Priority	23	Malicious Packets		8679567514	
Low Priority	33	True Negatives Passed		12230	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		13	
ML Metrics	Value	SYN Flood		19	
False Positives	78	UDP Flood		11	
FP Rate	0.58%	Uncertain		13	
IF Precision	96.60%				
IF Recall	80.72%	Action		Count	
IF F1-Score	87.95%	Quarantined		10	
IF Accuracy	96.23%	Time Ban		10	
		Blackhole		0	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	96.60%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	80.72%	Share of actual attacks successfully flagged
F1-Score	87.95%	Balanced measure combining Precision and Recall
Accuracy	96.23%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.58%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	19.28%	Actual attacks that went undetected
True Positive Rate (TPR)	80.72%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.42%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 1963	FN 469
Actual: Normal	FP 69	TN 11761

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.97%	Balanced measure combining Precision and Recall
Accuracy	99.95%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.45%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.55%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	1072	0	0
Act: ICMP	0	123	0
Act: UDP	0	0	768

2c. Hybrid Cascaded

Metric	Value	Description
Precision	96.60%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	80.72%	Share of actual attacks successfully flagged
F1-Score	87.95%	Balanced measure combining Precision and Recall
Accuracy	96.23%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.58%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	19.28%	Actual attacks that went undetected
True Positive Rate (TPR)	80.72%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.42%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 1963	FN 469
Actual: Normal	FP 69	TN 11761

2d. Response Latency

Metric	Value	Description
Detection Time	488.5 ms	Average time taken to detect an attack
Mitigation Response Time	16.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	4.66%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	8.37%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.26%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.23	4	5	2026-09-25 13:23:20	2026-09-25 13:29:28
10.0.0.16	4	3	2026-09-25 13:23:20	2026-09-25 13:29:16
10.0.0.17	4	3	2026-09-25 13:23:20	2026-09-25 13:29:17
10.0.0.18	3	3	2026-09-25 13:23:20	2026-09-25 13:26:30
10.0.0.19	3	3	2026-09-25 13:23:20	2026-09-25 13:26:30
10.0.0.20	3	3	2026-09-25 13:23:20	2026-09-25 13:29:17
10.0.0.21	3	3	2026-09-25 13:23:21	2026-09-25 13:29:24
10.0.0.24	3	3	2026-09-25 13:23:20	2026-09-25 13:29:24

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.25	3	3	2026-09-25 13:23:20	2026-09-25 13:29:23
10.0.0.22	2	1	2026-09-25 13:27:07	2026-09-25 13:29:34
10.0.0.4	2	1	2026-09-25 13:27:56	2026-09-25 13:28:28

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 13:23:20	10.0.0.19	DDoS	SYN Flood	90.1%	Critical	Quarantined
2026-09-25 13:23:20	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Quarantined
2026-09-25 13:23:20	10.0.0.24	DDoS	ICMP Flood	91.1%	High	Time Ban (1m)
2026-09-25 13:23:20	10.0.0.24	DDoS	ICMP Flood	91.1%	High	Time Ban
2026-09-25 13:23:20	10.0.0.25	DDoS	ICMP Flood	91.1%	High	Time Ban (1m)
2026-09-25 13:23:20	10.0.0.25	DDoS	ICMP Flood	91.1%	High	Time Ban
2026-09-25 13:23:20	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Quarantined
2026-09-25 13:23:20	10.0.0.20	DDoS	UDP Flood	76.4%	High	Time Ban (1m)
2026-09-25 13:23:20	10.0.0.20	DDoS	UDP Flood	76.4%	High	Time Ban
2026-09-25 13:23:20	10.0.0.23	DDoS	ICMP Flood	91.5%	High	Time Ban (1m)
2026-09-25 13:23:20	10.0.0.23	DDoS	ICMP Flood	91.5%	High	Time Ban
2026-09-25 13:23:21	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (1m)
2026-09-25 13:23:21	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-25 13:23:30	10.0.0.19	DDoS	SYN Flood	90.4%	Critical	Time Ban (1m)
2026-09-25 13:23:30	10.0.0.18	DDoS	SYN Flood	91.2%	High	Time Ban (1m)
2026-09-25 13:23:31	10.0.0.19	DDoS	SYN Flood	90.4%	Critical	Time Ban
2026-09-25 13:23:31	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Time Ban
2026-09-25 13:23:31	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (1m)
2026-09-25 13:23:31	10.0.0.16	DDoS	SYN Flood	90.4%	Critical	Time Ban (1m)
2026-09-25 13:24:20	10.0.0.24	DDoS	ICMP Flood	91.1%	High	Time Ban (2m)
2026-09-25 13:24:20	10.0.0.25	DDoS	ICMP Flood	91.1%	High	Time Ban (2m)
2026-09-25 13:24:21	10.0.0.20	DDoS	UDP Flood	76.4%	High	Time Ban (2m)
2026-09-25 13:24:21	10.0.0.23	DDoS	ICMP Flood	91.5%	High	Time Ban (2m)
2026-09-25 13:24:21	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (2m)
2026-09-25 13:24:30	10.0.0.19	DDoS	SYN Flood	91.9%	Critical	Time Ban (2m)
2026-09-25 13:24:30	10.0.0.18	DDoS	SYN Flood	91.8%	High	Time Ban (2m)
2026-09-25 13:24:31	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (2m)
2026-09-25 13:24:31	10.0.0.16	DDoS	SYN Flood	90.4%	Critical	Time Ban (2m)
2026-09-25 13:26:20	10.0.0.24	DDoS	ICMP Flood	91.1%	High	Time Ban (5m)
2026-09-25 13:26:20	10.0.0.25	DDoS	ICMP Flood	91.1%	High	Time Ban (5m)
2026-09-25 13:26:21	10.0.0.20	DDoS	UDP Flood	76.4%	High	Time Ban (5m)
2026-09-25 13:26:21	10.0.0.23	DDoS	ICMP Flood	91.5%	High	Time Ban (5m)
2026-09-25 13:26:21	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (5m)
2026-09-25 13:26:30	10.0.0.19	DDoS	SYN Flood	91.9%	Critical	Time Ban (5m)
2026-09-25 13:26:30	10.0.0.18	DDoS	SYN Flood	91.8%	Critical	Time Ban (5m)
2026-09-25 13:26:31	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (5m)
2026-09-25 13:26:31	10.0.0.16	DDoS	SYN Flood	90.4%	Critical	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 13:26:37	10.0.0.16	DDoS	SYN Flood	90.4%	Critical	Time Ban
2026-09-25 13:26:37	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban
2026-09-25 13:27:07	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:27:18	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Time Ban (1m)
2026-09-25 13:27:18	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Time Ban
2026-09-25 13:27:42	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-25 13:28:07	10.0.0.4	DDoS	Uncertain	59.0%	Low	Released
2026-09-25 13:28:08	10.0.0.20	DDoS	Uncertain	67.2%	Medium	Quarantined
2026-09-25 13:28:08	10.0.0.21	DDoS	Uncertain	67.5%	Medium	Quarantined
2026-09-25 13:29:06	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:29:07	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:29:12	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:29:13	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:29:18	10.0.0.23	DDoS	Uncertain	-	Low	Released
2026-09-25 13:29:23	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 13:29:24	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 13:29:24	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-25 13:29:28	10.0.0.23	DDoS	ICMP Flood	95.8%	Critical	Time Ban (10m)
2026-09-25 13:29:34	10.0.0.22	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 13:29:34	Attack Stopped
10.0.0.23	ICMP Flood	0.6422	95.8%	Critical	Phase 2	5	10s	2026-09-25 13:29:28	Attack persisted - escalating to time ban
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 13:29:24	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:29:24	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:29:23	Attack Stopped
10.0.0.23	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:29:18	Attack Stopped
10.0.0.17	SYN Flood	0.6456	89.7%	Critical	Phase 2	1	10s	2026-09-25 13:29:17	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6382	77.5%	High	Phase 2	1	11s	2026-09-25 13:29:17	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6444	93.1%	Critical	Phase 2	1	10s	2026-09-25 13:29:16	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.4	Uncertain	0.6114	59.0%	Low	Phase 1	1	10s	2026-09-25 13:28:28	Attack Stopped
10.0.0.4	Uncertain	0.6118	59.0%	Low	Phase 1	1	11s	2026-09-25 13:28:07	Attack Stopped
10.0.0.22	UDP Flood	0.6402	86.3%	Critical	Phase 2	1	11s	2026-09-25 13:27:18	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6410	90.4%	Critical	Phase 2	3	3m 11s	2026-09-25 13:26:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6414	92.6%	Critical	Phase 2	3	3m 11s	2026-09-25 13:26:31	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.5914	91.8%	Critical	Phase 2	3	3m 10s	2026-09-25 13:26:30	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6188	91.9%	Critical	Phase 2	3	3m 10s	2026-09-25 13:26:30	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6112	78.1%	High	Phase 2	3	3m 0s	2026-09-25 13:26:21	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6469	91.5%	High	Phase 2	3	3m 1s	2026-09-25 13:26:21	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.4%	High	Phase 2	3	3m 1s	2026-09-25 13:26:21	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6456	91.1%	High	Phase 2	3	3m 0s	2026-09-25 13:26:20	Escalated to Time Ban
10.0.0.24	ICMP Flood	0.6465	91.1%	High	Phase 2	3	3m 0s	2026-09-25 13:26:20	Escalated to Time Ban
10.0.0.16	SYN Flood	0.6410	90.4%	Critical	Phase 2	2	1m 11s	2026-09-25 13:24:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6414	92.6%	Critical	Phase 2	2	1m 11s	2026-09-25 13:24:31	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6146	91.8%	High	Phase 2	2	1m 10s	2026-09-25 13:24:30	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6226	91.9%	Critical	Phase 2	2	1m 10s	2026-09-25 13:24:30	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6112	78.1%	High	Phase 2	2	1m 0s	2026-09-25 13:24:21	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6469	91.5%	High	Phase 2	2	1m 1s	2026-09-25 13:24:21	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6452	76.4%	High	Phase 2	2	1m 1s	2026-09-25 13:24:21	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6456	91.1%	High	Phase 2	2	1m 0s	2026-09-25 13:24:20	Escalated to Time Ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.24	ICMP Flood	0.6465	91.1%	High	Phase 2	2	1m 0s	2026-09-25 13:24:20	Escalated to Time Ban
10.0.0.16	SYN Flood	0.6410	90.4%	Critical	Phase 2	1	11s	2026-09-25 13:23:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6414	92.6%	Critical	Phase 2	1	11s	2026-09-25 13:23:31	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6238	91.2%	High	Phase 2	1	10s	2026-09-25 13:23:30	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6376	90.4%	Critical	Phase 2	1	10s	2026-09-25 13:23:30	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE